

Nikolaos Goutzoulis

EDUCATION Phone: (646)-957-0924 | Email: ng2985@columbia.edu | LinkedIn: Nikos Goutzoulis

Columbia University

Computer Science MS

• GPA: 3.54/4.00

• Relevant Courses: Unsupervised Learning, Advanced Deep Learning, Advanced Big Data and AI

Aug 2024 - Dec 2025

New York, NY

National Technical University of Athens

Electrical and Computer Engineering - Integrated MEng

• Grade: 8.7/10 (top 5% of 400 students)

• Relevant Courses: Deep Learning, Databases, Advanced Algorithms

Jul 2024

Athens, GR

PROFESSIONAL EXPERIENCE

Ripple

Blockchain Data Scientist Intern

• Productionized a **Databricks ETL pipeline using SQL** to calculate DeFi metrics from raw, on-chain data. The analysis helped the DeFi team measure the ecosystem's health and make more informed decisions.

• Owned a Gemini 2.5 + n8n multi-agent flow for XRPL DEX tasks, containerized and orchestrated via **Kubernetes** for high availability, reduced clicks by 4x for non-technical users and raised task success to 95%.

• Developed an **end-to-end data pipeline** using **Python, PySpark, and the Fireblocks API** to automate the calculation of rolling asset balances, eliminating 100s of spreadsheets and saving about 10 hrs/week for Treasury.

May 2025 - Aug 2025

New York, NY

Columbia University

Course Assistant - Analysis of Algorithms

• Guided 150+ students in mastering data structures, **dynamic and linear programming**, and **nonlinear optimization** techniques, created 10+ problem walkthroughs and a DP cheat-sheet adopted by the course.

Jan 2025 - May 2025

New York, NY

Graduate Research Assistant – Software Systems Lab

• Collaborated with Meta researchers through **Git** to create a **browser API** in **C++** for Chrome by measuring user engagement without cross-site tracking and enabling secure attribution and analytics.

Sep 2024 - Apr 2025

EmDoT S.A.

Machine Learning Engineer - Thesis Collaboration

• Built deep learning models with **PyTorch** using **CUDA**, and machine learning models such as random forests with **Sklearn**, boosting prediction accuracy by 15% for resource allocation KPIs in an insurance company.

• Optimized model performance by 20% and accuracy by performing **feature engineering**, incorporating parameters such as time of day and custom metrics.

• Engaged with cross-functional teams, directly **interacted with customers** and **deployed ML models** in industrial use.

Oct 2023 - Jul 2024

Athens, GR

Data Science Intern

• Delivered algorithms for smoothing Bin Fill Level data, using **SQL**, **Tableau**, and statistical models like **Prophet** and **SARIMAX** to predict the trend of the data. Reduced noise in data by 20% while maintaining 95% of original signal.

Jul 2023 - Sep 2023

PROJECTS

NutritionAI: LLM Nutritional Personalized Assistant

Columbia University

• Preprocessed a conversational and nutritional dataset with **Python scripts** and fine tuned **Llama3 8B** on AWS's NVIDIA T4 GPU. The natural language processing project produced personalized nutritional programs for each user.

Jan 2025 - Mar 2025

Explainable Thyroid-Nodule Classification with Curriculum-Trained ResNet-18

Columbia University

• Built a CAD pipeline that fine-tunes ResNet-18 with **Adaptive Curriculum Loss**, achieving 0.91 AUC on the 3 k-image TN3K ultrasound dataset. Integrated real-time **Grad-CAM visualisations** and ROI cropping to expose and correct spurious model attention, improving both diagnostic accuracy and interpretability for clinicians.

Jan 2025 - Mar 2025

Evaluating LLM Resilience to Adversarial Data Poisoning

Columbia University

• Designed a method to assess the resilience of **RoBERTa LLM** in sentiment analysis against adversarial data poisoning, which revealed a model accuracy drop of up to 15%.

Sep 2024 - Dec 2024

• Generated text embeddings using BERT and performed **dimensionality reduction** within **Databricks** to facilitate sentiment analysis.

SKILLS

Programming Languages: Python, C++, JavaScript, Java, Matlab, **SQL**, Rust, R

ML Frameworks: Tensorflow, CUDA, **PyTorch**, Keras, **Scikit-learn**, Transformers, XGBoost

Cloud, Development and Other Tools: GCP, Databricks, Docker, Kubernetes, PowerBI, **PySpark**, API design, Visual Studio, **Matplotlib**, **Git**, Tableau, React, Node.js